

LOGISTIC REGRESSION PROJECT REPORT

Project: Diabetes Prediction using Kaggle Data
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Objective Predict Outcome: 0 = No diabetes, 1 = Diabetes.	Dataset Pima Indians Diabetes Database from Kaggle; file: diabetes.csv (768 rows, 8 inputs).	Model Median imputation + StandardScaler + balanced Logistic Regression.
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Dataset and Workflow

Inputs: Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, and Age.

Workflow: load Kaggle CSV → inspect data → replace invalid values → EDA → stratified 80/20 train-test split → median imputation → scaling → train Logistic Regression → evaluate → interpret coefficients.

Evaluation Metrics

Metric	Result	Meaning
Accuracy	0.721	Overall correct predictions
Precision	0.587	Reliability of positive predictions
Recall	0.685	Actual positive cases detected
F1-score	0.632	Balance of precision and recall
ROC-AUC	0.811	Overall class-separation ability

Confusion Matrix

	Predicted 0	Predicted 1
Actual 0	TN = 74	FP = 26
Actual 1	FN = 17	TP = 37

Key Findings

The model obtains a ROC-AUC of about 0.811 and detects approximately 68.5% of the positive cases (37 of 54) in the fixed test split of 154 patients. Glucose (1.19) and BMI (0.72) have the strongest positive model coefficients. Recall is especially important in a screening-style classification task because false negatives represent missed positive cases.

Limitation: This small historical dataset represents a specific population. The model is suitable for learning classification concepts, not for medical diagnosis.

Final Conclusion

In this project I used Logistic Regression to predict whether a person has diabetes. Before building the model I found that the Insulin and SkinThickness columns had filled-in values that depended on the Outcome, so I treated them as missing and filled them with the training median. The final model gave an accuracy of 0.721 and a recall of 0.685, which means it found most of the diabetic patients, but its precision of 0.587 shows that it also gives some false alarms (26 in the test set). I think this is acceptable for a screening-style task, since missing a diabetic patient is worse than doing one extra test. Glucose and BMI affected the prediction the most. The dataset is small, so the model is only for learning and not for real diagnosis.